

Khoi Nguyen

PhD in Computer Science and Artificial Intelligence

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EMPLOYMENT

Apr 2025 – Now

Staff Research Scientist at Qualcomm AI Research

- Conduct research in Computer Vision, Generative AI, and Robotics
- Publish papers at top-tier AI conferences such as CVPR, ICCV, ECCV, and NeurIPS
- Build efficient Video/3D/4D World Models and data generation system for Robot Learning and Mobile Applications.

Jul 2021 – Mar 2025

Research Scientist at VinAI Research

- Conduct cutting-edge research in Computer Vision, including Perception AI (detection, segmentation, tracking) and Generative AI (Image, Video, 3D Model Synthesis)
- Publish papers at top-tier CV conferences such as CVPR, ICCV, ECCV, and NeurIPS
- Lead a product to build a visual perception system for an autonomous driving agent

EDUCATION

Sep 2017 – Jun 2021

Doctor of Philosophy in Computer Science

Oregon State University, Corvallis, OR, USA, advisor: Prof. Sinisa Todorovic

- Thesis: "Part-based and Uncertainty-Aware Few-shot Object Segmentation in Images"

Sep 2015 – Sep 2017

Master of Science in Computer Science

Oregon State University, Corvallis, OR, USA, advisor: Prof. Sinisa Todorovic

- Thesis: "Relational Networks for Visual Relationship Detection"

RESEARCH EXPERTISE

Research Interests

- **Reinforcement Learning and World Models** for Robot Learning
- **3D/4D Reconstruction, Understanding, and Generation**
- **Deep Generative Models** (diffusion, flow-matching and autoregressive models) for Visual Generation and Editing

Technical Skills

- PyTorch, Python, C++, Diffusion Models, Auto-regressive Generative Models

PUBLICATIONS

Conferences
[Khoi Nguyen](#): Myself

- [CVPR'26b] Duc Vu*, Kien Nguyen*, Trong-Tung Nguyen*, Ngan Nguyen*, Phong Nguyen, [Khoi Nguyen](#), Cuong Pham, Anh Tran **InverFill: One-Step Inversion for Enhanced Few-Step Diffusion Inpainting**
- [CVPR'26 (Findings)] Huy Duong*, Trong-Tung Nguyen*, Cuong Pham, Anh Tran, [Khoi Nguyen](#), Minh Hoai **SwiftPie: Lightning-fast Subject-driven Image Personalization via One step Diffusion**
- [ICCV'25b] Viet Van Nguyen*, Anh Nguyen*, Trung Tuan Dao, [Khoi Nguyen](#), Cuong Pham, Toan Tran, Anh Tuan Tran **Supercharged One-step Text-to-Image Diffusion Models with Negative Prompts**
- [ICCV'25a] Quang-Binh Nguyen, Minh Luu, Quang Nguyen, Anh Tuan Tran, [Khoi Nguyen](#) **CSD-VAR: Content-Style Decomposition in Visual Autoregressive Models**
- [ICCV'25 (Workshops)] Phuc D.A. Nguyen*, Minh Luu*, Anh Tran, Cuong Pham, [Khoi Nguyen](#), **Open-Ended 3D Point Cloud Instance Segmentation**
- [CVPR'25c] Trong-Tung Nguyen, Quang Nguyen, [Khoi Nguyen](#), Anh Tran, Cuong Pham, **SwiftEdit: Lightning Fast Text-Guided Image Editing via One-Step Diffusion**
- [CVPR'25b] Hai Pham, Tung Do, Phong Nguyen, Son Hua, [Khoi Nguyen](#), Rang Nguyen, **SharpDepth: Sharpening Metric Depth Predictions Using Diffusion Distillation**
- [CVPR'25a] Phuc Nguyen*, Minh Luu*, Anh Tran, Cuong Pham, [Khoi Nguyen](#), **Any3DIS: Class-Agnostic 3D Instance Segmentation by 2D Mask Tracking**
- [AAAI'25b] Hung Nguyen*, Quang Qui-Vinh Nguyen*, [Khoi Nguyen](#), Rang Nguyen, **SwiftTry: Fast and Consistent Video Virtual Try-On with Diffusion Models**
- [AAAI'25a] Duc-Hai Pham, Duc Dung Nguyen, Hoang-Anh Pham, Ho Lai Tuan, Phong Ha Nguyen, [Khoi Nguyen](#), Rang Nguyen, **Semi-supervised 3D Semantic Scene Completion with 2D Vision Foundation Model Guidance**

- [ECCV'24b] Uy Dieu Tran*, Minh Luu*, Phong Nguyen, [Khoi Nguyen](#), Binh-Son Hua, **Diverse Text-to-3D Synthesis with Augmented Text Embedding**
 - [ECCV'24a] Trung Dao, Thanh Le, Duc Vu, Thuan Nguyen, [Khoi Nguyen](#), Cuong Pham, Anh Tran, **SBv2: Make Your One-step Diffusion Model Better Than Its Teacher**
 - [CVPR'24] Phuc DA Nguyen*, Tuan Duc Ngo*, Chuang Gan, Evangelos Kalogerakis, Anh Tran, Cuong Pham, [Khoi Nguyen](#), **Open3DIS: Open-vocabulary 3D Instance Segmentation with 2D Mask Guidance**
 - [WACV'24] Chau Pham*, Truong Vu*, [Khoi Nguyen](#), **LP-OVOD: Open-Vocabulary Object Detection by Linear Probing**
 - [NeurIPS'23] Quang Nguyen*, Truong Vu*, Anh Tran, [Khoi Nguyen](#), **Dataset-Diffusion: Diffusion-based Synthetic Data Generation for Pixel-Level Semantic Segmentation**
 - [ICCV'23] Tuan Ngo, Binh-Son Hua, [Khoi Nguyen](#), **GaPro: Box-Supervised 3D Point Cloud Instance Segmentation Using Gaussian Processes as Pseudo Labels**
 - [CVPR'23] Tuan Ngo, Binh-Son Hua, [Khoi Nguyen](#), **ISBNet: a 3D Point Cloud Instance Segmentation Network with Instance-aware Sampling and Box-aware Dynamic Convolution**
 - [WACV'23] Hue Nguyen, Diep Tran, [Khoi Nguyen](#), Rang Nguyen, **PSENet: Progressive Self-Enhancement Network for Unsupervised Extreme-Light Image Enhancement , (The Best Paper - Honorable Mention Award!)**
 - [ECCV'22c] Tuan Ngo, [Khoi Nguyen](#), **Geodesic-Former: a Geodesic-Guided Few-shot 3D Point Cloud Instance Segmenter**
 - [ECCV'22b] Thanh Nguyen*, Chau Pham*, [Khoi Nguyen](#), Minh Hoai, **Few-shot Object Counting and Detection**
 - [ECCV'22a] Khoi D. Nguyen, Quoc-Huy Tran, [Khoi Nguyen](#), Binh-Son Hua, Rang Nguyen, **Inductive and Transductive Few-Shot Video Classification via Appearance and Temporal Alignments**
 - [CVPR'22] [Khoi Nguyen](#), Sinisa Todorovic, **iFS-RCNN: An Incremental Few-shot Instance Segmenter**
 - [NeurIPS'21] Duong Le*, Khoi D. Nguyen*, [Khoi Nguyen](#), Quoc-Huy Tran, Rang Nguyen, Binh-Son Hua, **POODLE: Improving Few-shot Learning via Penalizing Out-of-Distribution Samples**
 - [ICCV'21] [Khoi Nguyen](#), Sinisa Todorovic, **A Weakly Supervised Amodal Segmenter with Boundary Uncertainty Estimation**
 - [CVPR'21] [Khoi Nguyen](#), Sinisa Todorovic, **FAPIS: A Few-shot Anchor-free Part-based Instance Segmenter**
 - [ICPR'20] [Khoi Nguyen](#), Sinisa Todorovic, **A Self-supervised GAN for Unsupervised Few-shot Object Recognition**
 - [ICCV'19] [Khoi Nguyen](#), Sinisa Todorovic, **Feature Weighting and Boosting for Few-Shot Segmentation**
- Preprints
- [ArXiv'24c] Quang Nguyen, Truong Vu, Trong-Tung Nguyen, Yuxin Wen, Preston K Robinette, Taylor T Johnson, Tom Goldstein, Anh Tran, [Khoi Nguyen](#), **EditScout: Locating Forged Regions from Diffusion-based Edited Images with Multimodal LLM**
 - [ArXiv'24b] Uy Dieu Tran*, Minh Luu*, Phong Ha Nguyen, [Khoi Nguyen](#), Binh-Son Hua, **ModeDreamer: Mode Guiding Score Distillation for Text-to-3D Generation using Reference Image Prompts**
 - [arXiv'23] Quang Nguyen*, Truong Vu*, Cuong Pham, Anh Tran, [Khoi Nguyen](#), **Stable Messenger: Steganography for Message-Concealed Image Generation**
- Patents
- Tuan Trung Dao, Hoang Thuan Nguyen, Van Thanh, Le Hong Duc Vu, [Khoi Nguyen](#), Van Cuong PHAM, Tuan Anh Tran, Hai Hung BUI, **Text-to-image using low rank adaptation diffusion**, in *US Patent*, 2026
 - Ho Quang Nguyen, Tuan Truong Vu, Tuan Anh Tran, [Khoi Nguyen](#), Hai Hung BUI, **System and method for generating synthetic image and semantic segmentation mask thereof**, in *US Patent*, 2026
 - Thi Hue Nguyen, Thi Ngoc Diep Tran, Cong Thanh Tran, [Khoi Nguyen](#), Ho Man Rang Nguyen, Hai Hung Bui, **Method and apparatus for extreme-light image enhancement**, in *US Patent*, 2024
 - [Khoi Nguyen](#), Maneesh Singh, **Computer Vision Systems and Methods for Information Extraction from Text Images Using Evidence Grounding Techniques**, in *US Patent*, 2021

AWARDS/ACTIVITIES

- Awards
 - **Best Paper - Honorable Mention Award, WACV 2023**
 - Outstanding Reviewer Award, ECCV 2020
 - Vietnam Education Foundation (VEF) Fellowship (Cohort 2015)
- Area Chair
 - **NeurIPS** 2025, 2026; **ACCV**: 2024; **BMVC**: 2024, 2025;
- Reviewer
 - **CVPR**: 2021, 2022, 2023, 2024, 2025, 2026; **ICCV**: 2021, 2023, 2025; **ECCV**: 2020, 2022, 2024, 2026
 - **NeurIPS**: 2022, 2023, 2024; **ICLR**: 2022, 2023, 2024, 2025, 2026; **AAAI**: 2026
 - **Journals**: TPAMI, TIP

LEADERSHIP/MENTORSHIP

- Workshop Organizer
 - “SyntaGen: Harnessing Generative Models for Synthetic Visual Datasets”, CVPR 2024, 2025
- Team Lead
 - “Generative AI” research theme and Seminar Series Manager at VinAI Research
- Mentor
 - Mentoring 15+ AI residents to conduct cutting-edge research to publish 20+ papers on top-tier Computer Vision conferences and apply for 7+ prestigious PhD Programs.